



AI-DRIVEN

CYBER SECURITY

*EVERY CYBERATTACK TRAVELS THROUGH A NETWORK. IF YOU
DON'T UNDERSTAND NETWORKING AND OPERATING SYSTEMS,
YOU'LL NEVER TRULY UNDERSTAND CYBERSECURITY.*

WHAT IS A COMPUTER NETWORK?

- DEVICES CONNECTED TOGETHER
- COMMUNICATION BETWEEN SYSTEMS
- DATA SHARING
- INTERNET AS A GLOBAL NETWORK
- WIRED & WIRELESS NETWORKS

NETWORK COMPONENTS

- CLIENT
- SERVER
- ROUTER
- SWITCH
- MODEM
- ACCESS POINT
- FIREWALL

IP ADDRESS

- UNIQUE DEVICE IDENTIFIER
- IPV4 & IPV6
- PUBLIC IP
- PRIVATE IP
- STATIC VS DYNAMIC IP

MAC ADDRESS

- PHYSICAL HARDWARE ADDRESS
- UNIQUE NETWORK IDENTIFIER
- ASSIGNED BY MANUFACTURER
- WORKS ON LOCAL NETWORK

PORTS

- COMMUNICATION ENDPOINTS
- PORT NUMBERS
- OPEN VS CLOSED PORTS
- COMMON SERVICE PORTS

PROTOCOLS

- COMMUNICATION RULES
- HTTP
- HTTPS
- DNS
- DHCP
- FTP
- SSH
- TCP
- UDP

TCP VS UDP

- RELIABLE VS FAST
- CONNECTION-ORIENTED
- CONNECTIONLESS
- ERROR CHECKING
- REAL-TIME COMMUNICATION

DNS

- DOMAIN NAME RESOLUTION
- HUMAN-FRIENDLY NAMES
- IP LOOKUP
- INTERNET PHONEBOOK

HTTP VS HTTPS

- WEB COMMUNICATION
- ENCRYPTION
- SSL/TLS
- SECURE BROWSING

OPERATING SYSTEM

- MANAGES HARDWARE
- RUNS APPLICATIONS
- USER INTERFACE
- RESOURCE MANAGEMENT
- SECURITY CONTROLS

LINUX BASICS

- FILE SYSTEM
- ROOT USER
- TERMINAL
- PERMISSIONS
- PROCESSES

FILE PERMISSIONS

- READ (R)
- WRITE (W)
- EXECUTE (X)
- OWNER
- GROUP
- OTHERS

PROCESSES

- RUNNING PROGRAMS
- PROCESS ID (PID)
- BACKGROUND PROCESSES
- PROCESS MANAGEMENT

WHY ENGINEERS MUST LEARN NETWORKING & OS?

- UNDERSTAND ATTACKS
- ANALYZE TRAFFIC
- DETECT THREATS
- SECURE SYSTEMS
- TROUBLESHOOT ISSUES
- BUILD STRONG FOUNDATIONS